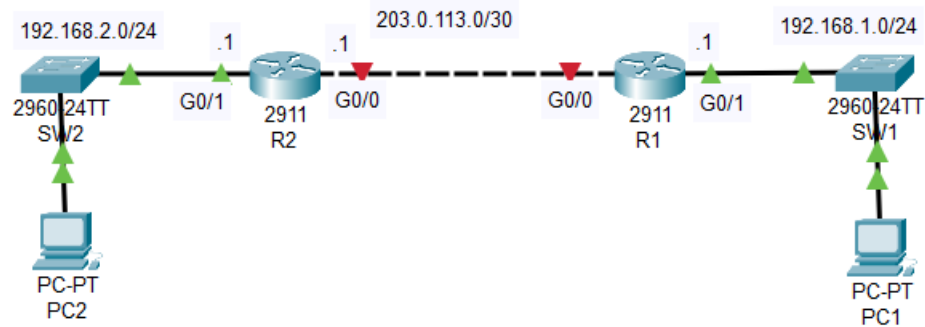


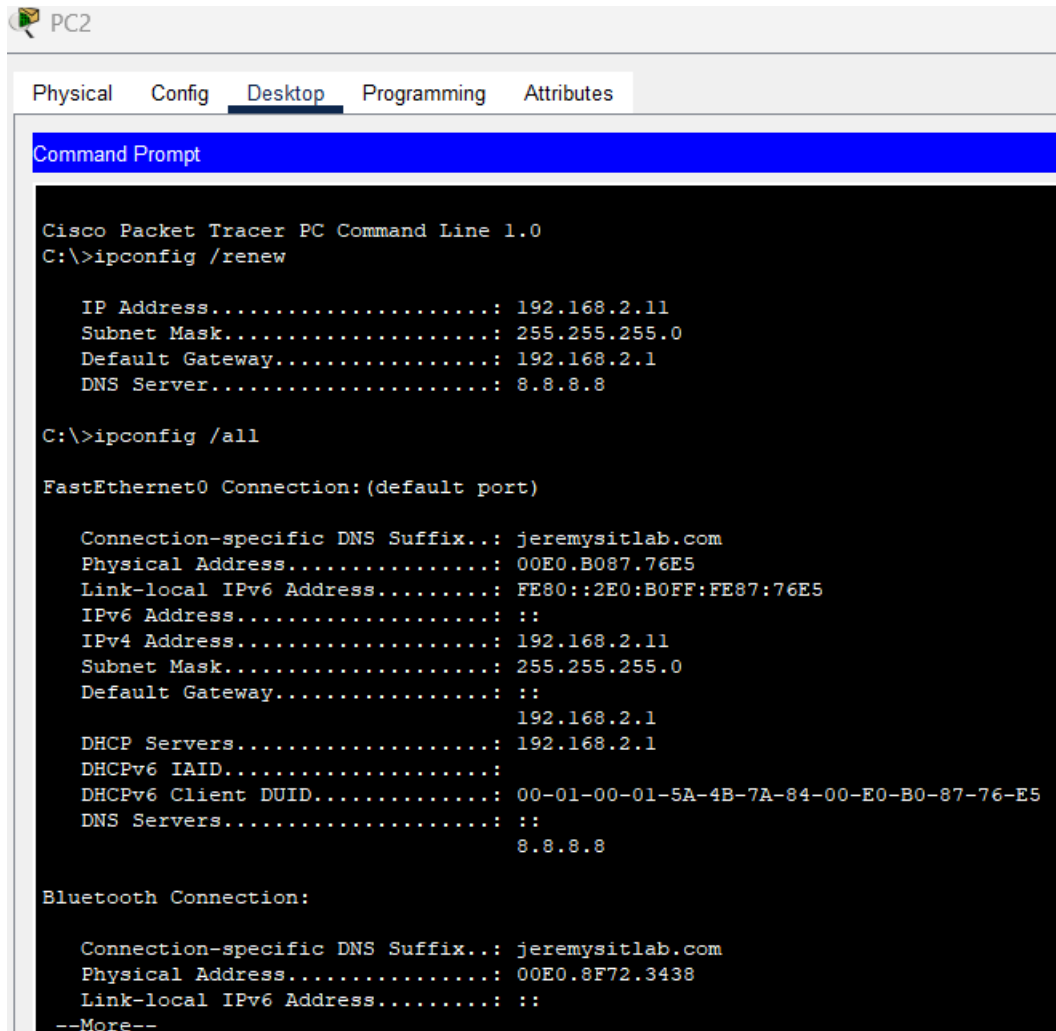


Topology

1. Configure the following DHCP pools on R2:

- **POOL1: 192.168.1.0/24 (reserve .1 to .10)**
DNS 8.8.8.8
Domain: jeremysitlab.com
Default Gateway: R1
- **POOL2: 192.168.2.0/24 (reserve .1 to .10)**
DNS 8.8.8.8
Domain: jeremysitlab.com
Default Gateway: R2
- **POOL3: 203.0.113.0/30 (reserve .1)**

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
R2(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.10
R2(config)#ip dhcp excluded-address 203.0.113.1
R2(config)#ip dhcp pool POOL1
R2(dhcp-config)#network 192.168.1.0 255.255.255.0
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#domain-name jeremysitlab.com
R2(dhcp-config)#default-router 192.168.1.1
R2(dhcp-config)#ip dhcp pool POOL2
R2(dhcp-config)#network 192.168.2.0 255.255.255.0
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#domain-name jeremysitlab.com
R2(dhcp-config)#default-router 192.168.2.1
R2(dhcp-config)#ip dhcp pool POOL3
R2(dhcp-config)#network 203.0.113.0 255.255.255.252
R2(dhcp-config)#do sh run | section dhcp
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp excluded-address 203.0.113.1
ip dhcp pool POOL1
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
domain-name jeremysitlab.com
ip dhcp pool POOL2
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
dns-server 8.8.8.8
domain-name jeremysitlab.com
ip dhcp pool POOL3
network 203.0.113.0 255.255.255.252
```



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /renew

    IP Address. . . . . : 192.168.2.11
    Subnet Mask. . . . . : 255.255.255.0
    Default Gateway. . . . . : 192.168.2.1
    DNS Server. . . . . : 8.8.8.8

C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix... : jeremysitlab.com
    Physical Address. . . . . : 00E0.B087.76E5
    Link-local IPv6 Address . . . . . : FE80::2E0:B0FF:FE87:76E5
    IPv6 Address. . . . . : ::
    IPv4 Address. . . . . : 192.168.2.11
    Subnet Mask. . . . . : 255.255.255.0
    Default Gateway. . . . . : ::
                                192.168.2.1
    DHCP Servers. . . . . : 192.168.2.1
    DHCPv6 IAID. . . . . :
    DHCPv6 Client DUID. . . . . : 00-01-00-01-5A-4B-7A-84-00-E0-B0-87-76-E5
    DNS Servers. . . . . : ::
                                8.8.8.8

Bluetooth Connection:

    Connection-specific DNS Suffix... : jeremysitlab.com
    Physical Address. . . . . : 00E0.8F72.3438
    Link-local IPv6 Address . . . . . : ::
    --More--
```

2. Configure R1's G0/0 interface as a DHCP client. What IP address did it configure?

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#
%DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/0 assigned DHCP address 203.0.113.2, mask
255.255.255.252, hostname R1
```

R1 will broadcast a DHCP Discover message to R2, which should reply with a DHCP offer, then R1 sends a DHCP Request, and R2 will send a DHCP Ack. As shown in the Syslog message, it was assigned the IP address of 203.0.113.2.

3. Configure R1 as a DHCP relay agent for the 192.168.1.0/24 subnet.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int g0/1
R1(config-if)#ip helper-address 203.0.113.1
```

4. Use the CLI of PC1 and PC2 to make them request an IP address from their DHCP server.

 PC1

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /renew
DHCP request failed.

C:\>ipconfig /renew

IP Address.....: 192.168.1.12
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.1.1
DNS Server.....: 8.8.8.8

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...: jeremysitlab.com
Physical Address.....: 0030.F238.8690
Link-local IPv6 Address.....: FE80::230:F2FF:FE38:8690
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.12
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
 192.168.1.1
DHCP Servers.....: 203.0.113.1
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-94-AE-8E-6C-00-30-F2-38-86-90
DNS Servers.....: ::
 8.8.8.8

Bluetooth Connection:

Connection-specific DNS Suffix...: jeremysitlab.com
Physical Address.....: 000A.F323.ACE5
Link-local IPv6 Address.....: ::
--More--